

2 Add a new loop

To make the clue the correct length once the secret word has been selected, use this simple loop. It counts how many letters are in the word and adds a question mark for each letter.

```
clue = []
index = 0
while index < len(secret_word):
    clue.append('?')
    index = index + 1
```

The `append()` function simply adds an item to the end of the list.

Make the ending smarter

At the moment, the game doesn't end until you type out the word in full. Let's make the code smarter so the game ends when you guess the last letter.

1 Make another variable

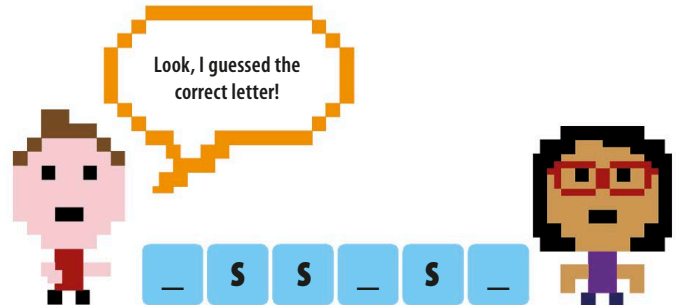
First create a variable to keep count of how many letters are unknown. Add this code above the `update_clue` function.

```
unknown_letters = len(secret_word)
```

At first all the letters are unknown.

2 Edit function

Next change the `update_clue()` function as shown below. Each time the player guesses a letter correctly, the program will now take away the number of times that letter appears in the secret word from `unknown_letters`.



```
def update_clue(guesses_letter, secret_word, clue, unknown_letters):
    index = 0
    while index < len(secret_word):
        if guesses_letter == secret_word[index]:
            clue[index] = guesses_letter
            unknown_letters = unknown_letters - 1
        index = index + 1

    return unknown_letters
```

Add this new parameter to the `update_clue` function.

The code subtracts 1 from `unknown_letters` each time a guessed letter appears in the word.

This line makes the function return the number of unknown letters.